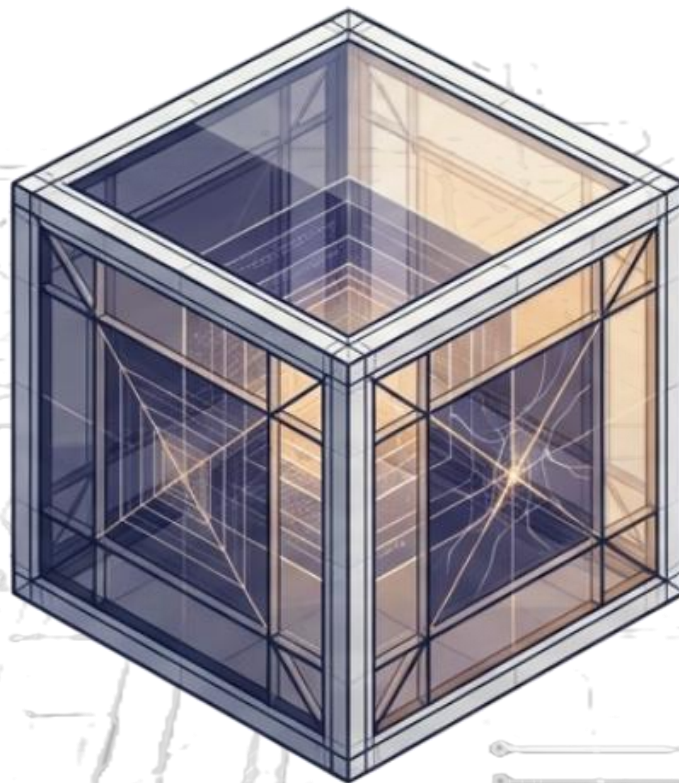


CNN FEATURE VISUALISATION AND DEEP DREAM

David Farrugia

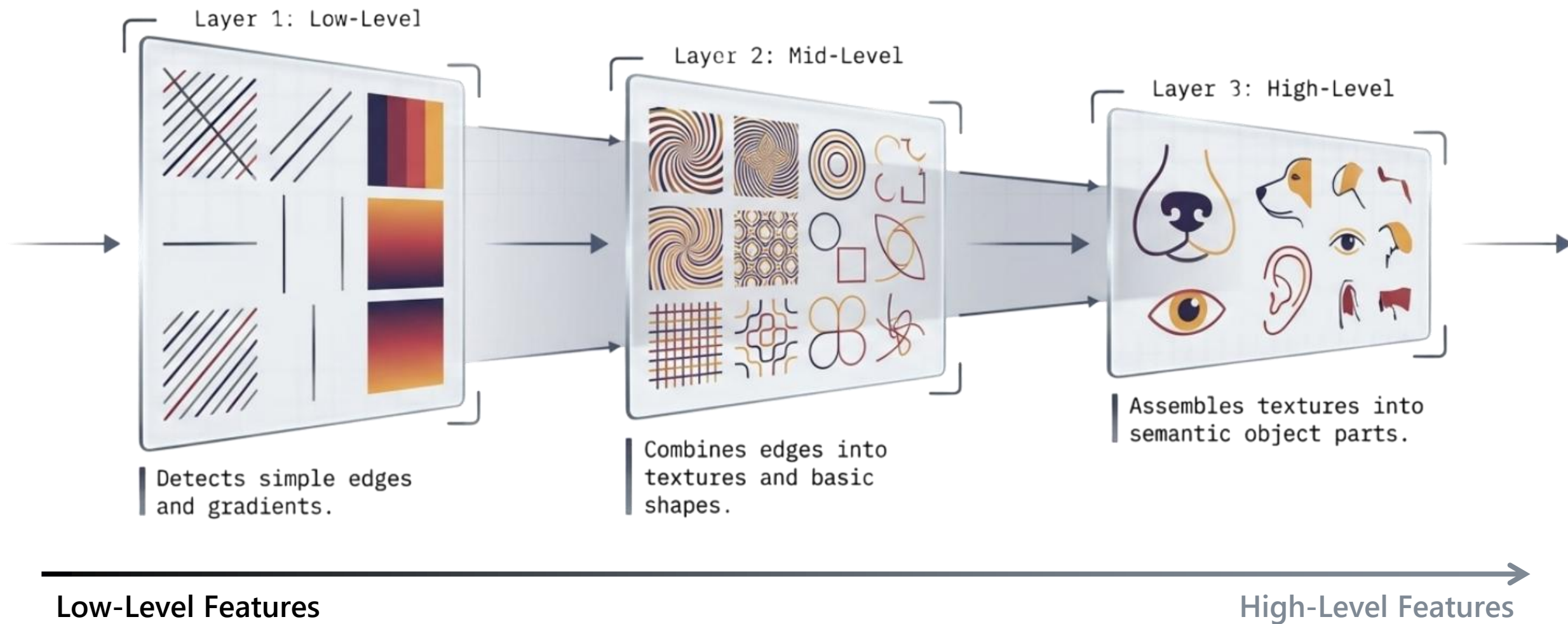
Why CNN Feature Visualisation Matters?



Prediction

Parrot
96%

CNN Feature Hierarchy



Activation Maximisation



The Goal is to find an input image that maximises the activation of a neuron or filter.



We aim to **maximise the activation** $a_i(x)$ of a chosen neuron i



Start of from a **random input image** x_0 (random noise tensor)



Use **gradient ascent** to iteratively update the image



The final image **reveals the patterns** that excite the neuron the most

$$x^* = \arg \max_x a_i(x)$$

Optimal images x that excite neuron i

$$x_{t+1} = x_t + \eta \frac{\partial a_i(x)}{\partial x}$$

Update rule using gradient ascent

Where:

x = input image (what we optimise)

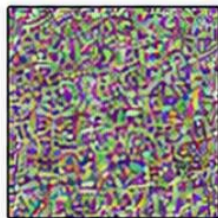
$a_i(x)$ = activation of neuron i for input x

η = step size (learning rate)

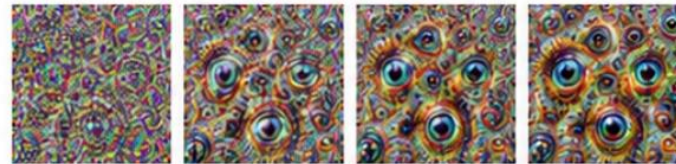
$\frac{\partial a_i(x)}{\partial x}$ = gradient of the activation with respect to the input

t = iteration step

1. Random Noise



2. Iterative Updates



.....

Gradient ascent updates

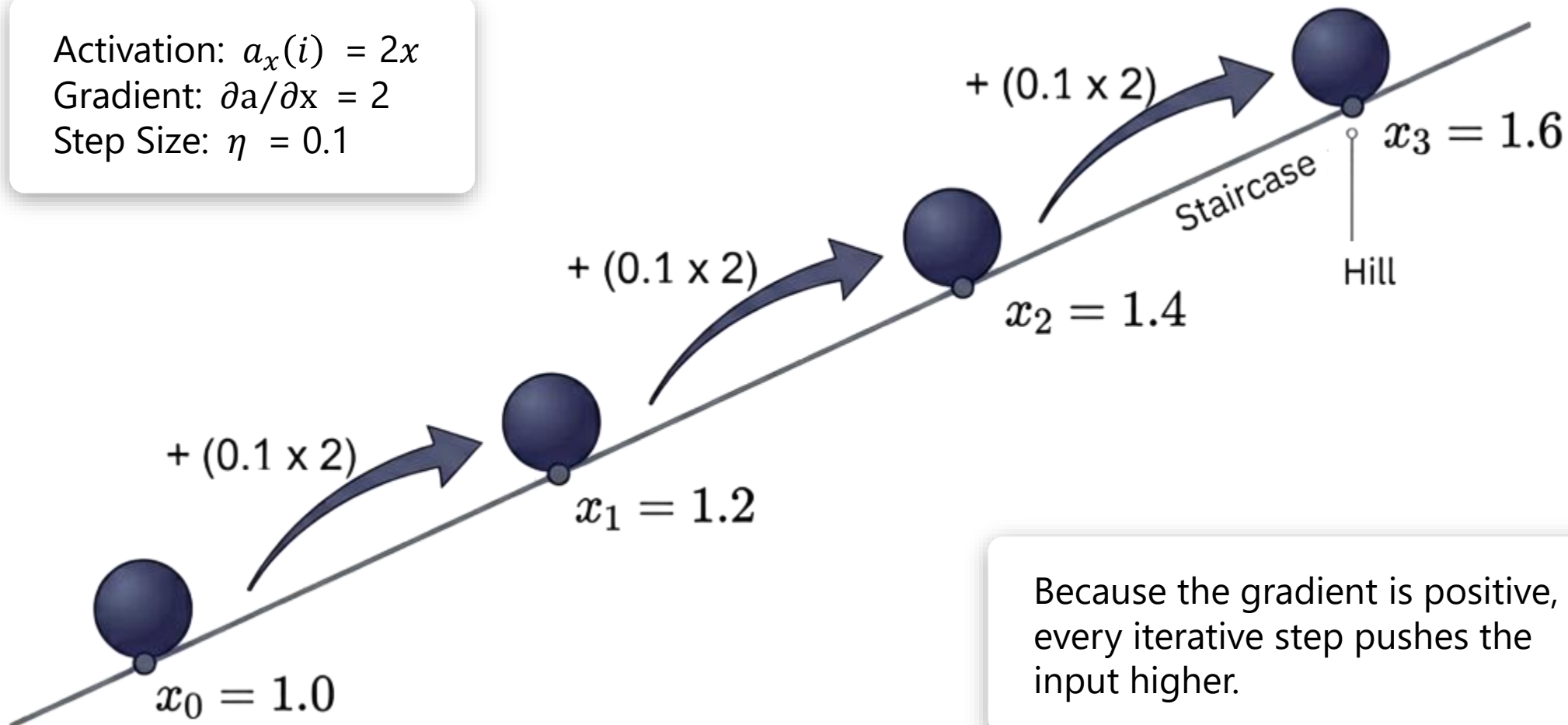
3. Final Pattern (Max Activation)



Image that maximally activates the selected neuron or filter.

Gradient Ascent in Action

Activation: $a_x(i) = 2x$
Gradient: $\partial a / \partial x = 2$
Step Size: $\eta = 0.1$



Because the gradient is positive, every iterative step pushes the input higher.

Regularisation Effects

$$x_{t+1} = x_t + \eta \left(\underbrace{\frac{\partial a_i(x)}{\partial x}}_{\text{Increases Activation}} - \lambda \underbrace{\frac{\partial R(x)}{\partial x}}_{\text{Regularisation (penalty)}} \right)$$

x : input image
 $a_i(x)$: activation of neuron i
 η : step size (learning rate)
 $R(x)$: regularisation function
 λ : regularisation strength



Activation Term $\frac{\partial a_i(x)}{\partial x}$

Pushes the image in the direction that **increases** the activation of neuron i .

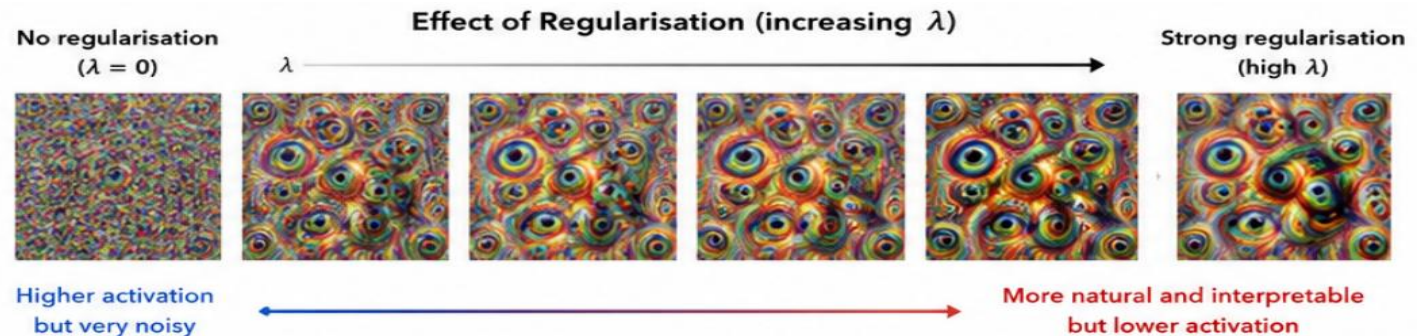


Regularisation Term $-\lambda \frac{\partial R(x)}{\partial x}$

Penalises undesirable properties (e.g. noise), encouraging natural-looking images.

Common Regularisers $R(x)$

-  **Total Variation (TV)**
reduces high-frequency noise
-  **L2 Norm**
Discourages extreme pixel values
-  **Gaussian Blur**
Suppresses high-frequency noise

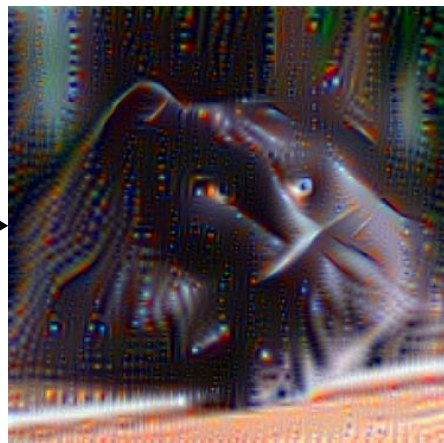
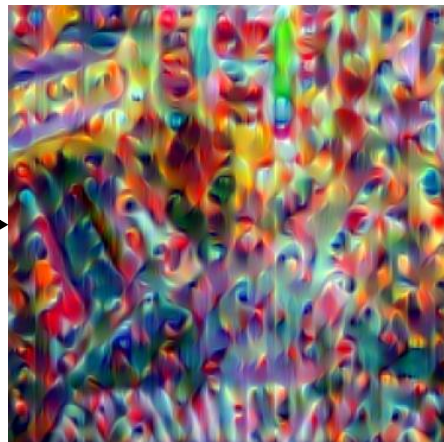


DeepDream Amplification

Original Images



DeepDream Output



$$x_{t+1} = x_t + \eta \frac{\partial}{\partial x} \left(\sum_{i \in \text{layer}} a_i(x) \right)$$



Starts from a real image

Instead of noise, DeepDream uses a real image as the starting point



Amplifies learned patterns

It enhances the patterns, textures and structures already present in the image



Optimises an entire layer

DeepDream optimises the activations of all neurons in a layer simultaneously, not just one



Different layers, different dreams

Earlier layers produce simpler textures and edges, while deeper layers create more complex object-like hallucinations



CONTROL PANEL

Single Mode

Compare Mode

METHOD A

1

Activation Maximization

METHOD B

1

DeepDream

CONV LAYER

1

Conv1

FILTER / CHANNEL

1

Filter 1

PARAMETERS

(AM) Step Size

0.001

(AM) L2 Regularisation

1

(DeepDream)

Octaves

2

(DeepDream) Octave Scale

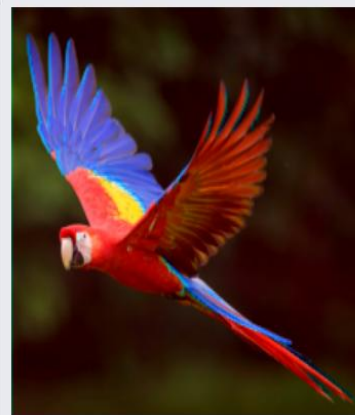
1.0

Reset Parameters

CNN ARCHITECTURE OVERVIEW



FEATURE VISUALIZATION (CONV1 - FILTER1)



Activation Maximisation



DeepDream

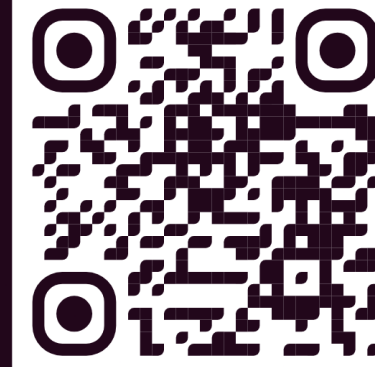
FEATURE MAPS (AM - CONV1)



INPUT IMAGES



SCAN ME



Limitations and Alternatives



Failure cases

- Repetitive textures or colour blobs
- DeepDream may hallucinate random patterns
- Outputs can be sensitive to parameter choices



When not to use

- For safety-critical decisions
- Poor choice when you need one clear explanation for a real prediction
- Not proof that the model understands the object



Alternatives

- Grad-CAM: highlights important image regions
- Saliency / Guided BP: pixel-level sensitivity
- Feature maps: inspect intermediate activations.

Take Away: Good for understanding CNNs, but not reliable for critical decisions

References Used

- [1] C. Olah, A. Mordvintsev, and L. Schubert, "Feature Visualization," Distill, vol. 2, no. 11, Nov. 2017, doi: <https://doi.org/10.23915/distill.00007>.
- [2] Mordvintsev et al. "Inceptionism: Going Deeper into Neural Networks," research.google. <https://research.google/blog/inceptionism-going-deeper-into-neural-networks/>
- [3] K. Simonyan, A. Vedaldi, and A. Zisserman, "Deep Inside Convolutional Networks: Visualising Image Classification Models and Saliency Maps," arXiv.org, 2013. <https://arxiv.org/abs/1312.6034>
- [4] A. Mahendran and A. Vedaldi, "Visualizing Deep Convolutional Neural Networks Using Natural Pre-images," International Journal of Computer Vision, vol. 120, no. 3, pp. 233–255, May 2016, doi: <https://arxiv.org/abs/1512.02017>.
- [5] A. Nguyen, J. Yosinski, and J. Clune, "Understanding Neural Networks via Feature Visualization: A survey," arXiv.org, 2019. <https://arxiv.org/abs/1904.08939>
- [6] M. D. Zeiler and R. Fergus, "Visualizing and Understanding Convolutional Networks," arXiv.org, 2013. <https://arxiv.org/abs/1311.2901>

THANK YOU

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[DavidF-22/ARI5118-DeepLearningCV Project](#)



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